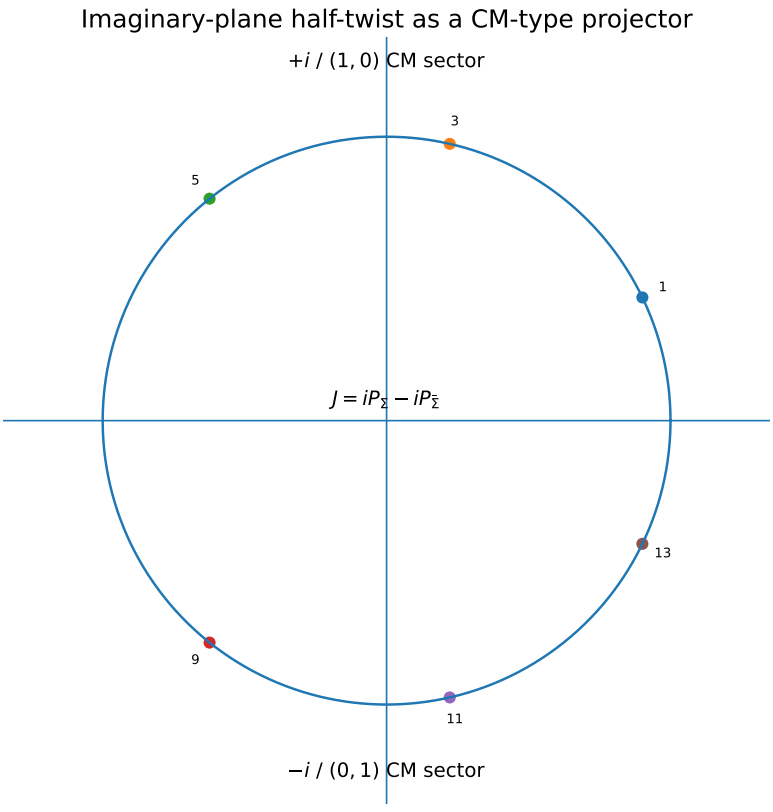
Complete Fermat Fourfold and CM Half-Twist Edition

Projector-Resolved Character Packet Calculus

Balanced Character Packets, Exact Conductors, Delsarte Transfers, Weighted Dyadic Refinement, Kummer Polyphase, and Imaginary CM Weights

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A constructive theorem–proof monograph on Fermat character packets, algebraic transfers, and CM half-twist carriers.

August 2026

Abstract

This monograph develops a constructive calculus for the primitive middle cohomology of the Fermat fourfold

$$X_m^4 = \{x_0^m + \cdots + x_5^m = 0\} \subset \mathbf{P}^5.$$

Diagonal symmetry resolves the cohomology into one-dimensional complex character lines and finite rational Galois packets. The central objects are balanced six-packets, exact conductor projectors, prime-local refinement laws, nonnegative packet identities, explicit monomial and incidence correspondences, and cyclotomic CM half-twists.

The main development includes the finite balanced-packet equation, centered Fourier coordinates, conductor descent and Möbius exact-conductor projectors, complementary-pair and saturated-join certificates, dual charges, curve and Delsarte channels, the exceptional degree-33 Klein–Delsarte transfer, infinite packet towers, odd prime-power classification, weighted dyadic reconstruction, a finite audit through degree 100, CRT–Kummer phase grammar, K4 and K23 reductions, a pure-imaginary cyclotomic selector, and explicit lower Fermat-curve CM carriers. Positive cycle frames and divisor-lowering transfers provide the synthesis layer used to reconstruct packet classes from explicit correspondences.

The constructive endpoint is a rank-one Chow correspondence from the selected lower Fermat-curve CM factor to each remaining K23 weight-one packet. The manuscript separates symbolic identities, finite exact certificates, correspondence-level theorems, published background, and open construction maps.

Purpose of this edition

This edition begins from the Fermat character decomposition and treats Hodge type as one address in the packet system. The primary question is constructive:

Given a rational Fermat character packet, what explicit cycle, transfer, or correspondence realizes it?

Published all-degree existence results are treated as background. The focus here is the explicit packet grammar, the prime-local decomposition, the CM half-twist carrier, and the algorithm that turns those data into cycle constructions.

Reader's map

The constructive pipeline is

$$\boxed{\begin{array}{c}\text{balanced six-packet} \longrightarrow \text{exact conductor} \longrightarrow \text{prime-local refinement} \\ \text{certificate channel} \longrightarrow \text{algebraic correspondence} \\ \text{rational packet reconstruction}\end{array}}$$

The main certificate channels are

pair, saturated join, curve, Delsarte, weighted dyadic, Kummer–Schur, CM half-twist.

Layer	Main object	Constructive question
Character address	$a = (a_0, \dots, a_5)$	Which rational Galois packet is selected?
Balance	$\ell_a(u) = 3$	Does every conjugate remain in the middle packet?
Conductor	$Q_{m,c}$	Which minimal modulus carries the packet?
Local phase	dyadic or mod-5 children	Which prime-local grammar applies?
Certificate	pair, SAT, curve, Delsarte, Kummer	Which explicit correspondence channel is available?
Transfer	$T = \sum_r B_r A_r$	Does the correspondence return the full packet?
CM carrier	$e_\Sigma H^1(F_d)$	Which lower curve carries the half-twist?
Reconstruction	nonzero projected cycle	How is the rational orbit recovered exactly?

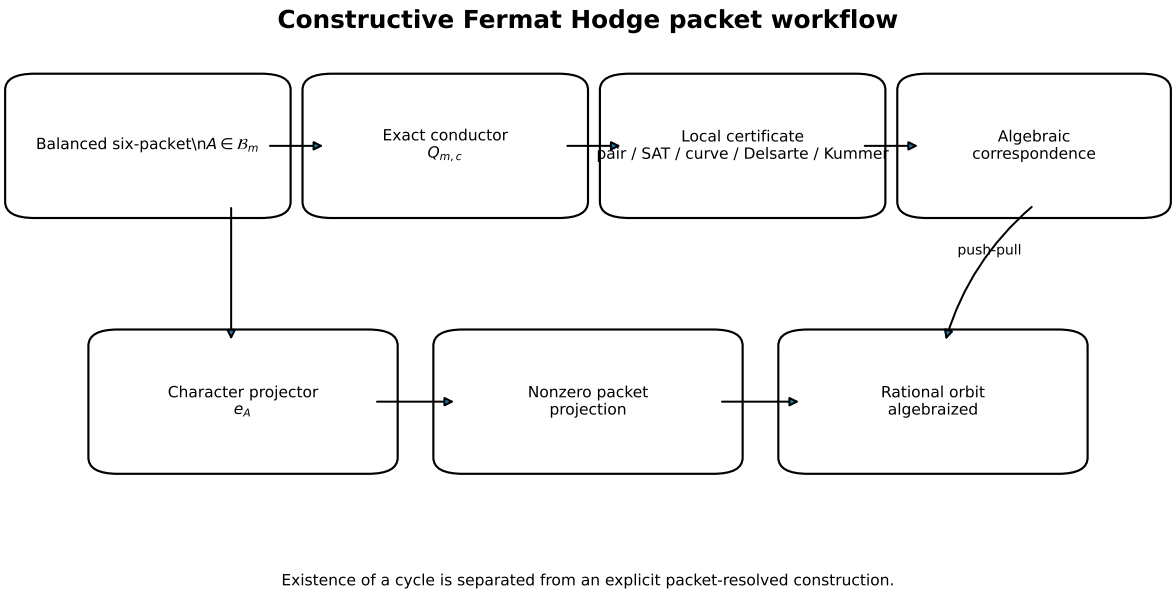


Figure 1: Constructive Fermat packet workflow from a balanced address to a nonzero packet-resolved cycle.

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Part I. Fermat Fourfold Character Geometry

Chapter 1

Program, objects, and constructive objective

For each integer $m \geq 3$, consider the smooth Fermat fourfold

$$X_m^4 = \{x_0^m + x_1^m + x_2^m + x_3^m + x_4^m + x_5^m = 0\} \subset \mathbf{P}^5.$$

The diagonal symmetry group resolves primitive middle cohomology into one-dimensional complex character lines. Rational packets are finite Galois orbits of these lines. This makes the geometry unusually explicit: each packet has a residue address, an exact conductor, a local prime profile, and a finite list of possible correspondence channels.

Constructive objective

For every rational character packet

$$W \subset H^4_{\mathrm{prim}}(X_m^4, \mathbf{Q})$$

of middle type, construct an explicit algebraic cycle or algebraic correspondence whose cohomology projection to W is nonzero.

The aim is not merely to know that some class exists. The aim is to produce:

1. a finite symbolic certificate identifying the packet
2. an explicit correspondence channel
3. a nonzero matrix coefficient or period certificate
4. a rational reconstruction of the full Galois orbit

The packet calculus uses Hodge level as a selection address, conductor as a polyphase depth, local prime data as a refinement grammar, and algebraic correspondences as analysis–synthesis maps. A packet may fail one grammar without being exceptional geometrically. The degree-33 construction is the central example: ordinary pair and saturation channels miss the packet, while a cross-family cubic transfer reaches it exactly.

Core distinction

$$\text{abstract existence} \quad \neq \quad \text{explicit packet-resolved construction.}$$

This book studies the second problem.

Chapter 2

Character packets and balanced residue equations

Let

$$U_m = (\mathbf{Z}/m\mathbf{Z})^\times$$

and let $\langle r \rangle_m \in \{1, \dots, m\}$ denote the least positive residue, with $\langle r \rangle_m = m$ when $m \mid r$. A primitive character address is a six-tuple

$$a = (a_0, \dots, a_5), \qquad 1 \le a_i < m, \qquad \sum_i a_i \equiv 0 \pmod{m}.$$

The Hodge level of the conjugate indexed by $u \in U_m$ is

$$\ell_a(u) = \frac{1}{m} \sum_{i=0}^{5} \langle u a_i \rangle_m.$$

The character line has type $(2,2)$ exactly when $\ell_a(1) = 3$. Its rational Galois orbit is a Hodge packet exactly when

$$\boxed{\ell_a(u) = 3 \qquad \forall u \in U_m.} \tag{B1}$$

To remove coordinate permutations, introduce the count vector

$$n(a) = (n_1, \dots, n_{m-1}) \in \mathbf{N}^{m-1}, \qquad n_r = \#\{i : a_i = r\}.$$

Let

$$R_m(u, r) = \langle ur \rangle_m.$$

Then the complete balanced six-packet space is the finite integer set

$$\boxed{\mathcal{B}_m = \left\{ n \in \mathbf{N}^{m-1} : \mathbf{1}^T n = 6, \quad R_m n = 3m\,\mathbf{1} \right\}.} \tag{B2}$$

The unit group permutes the residue coordinates. Rational packets correspond to U_m-orbits in $\mathcal{B}_m$.

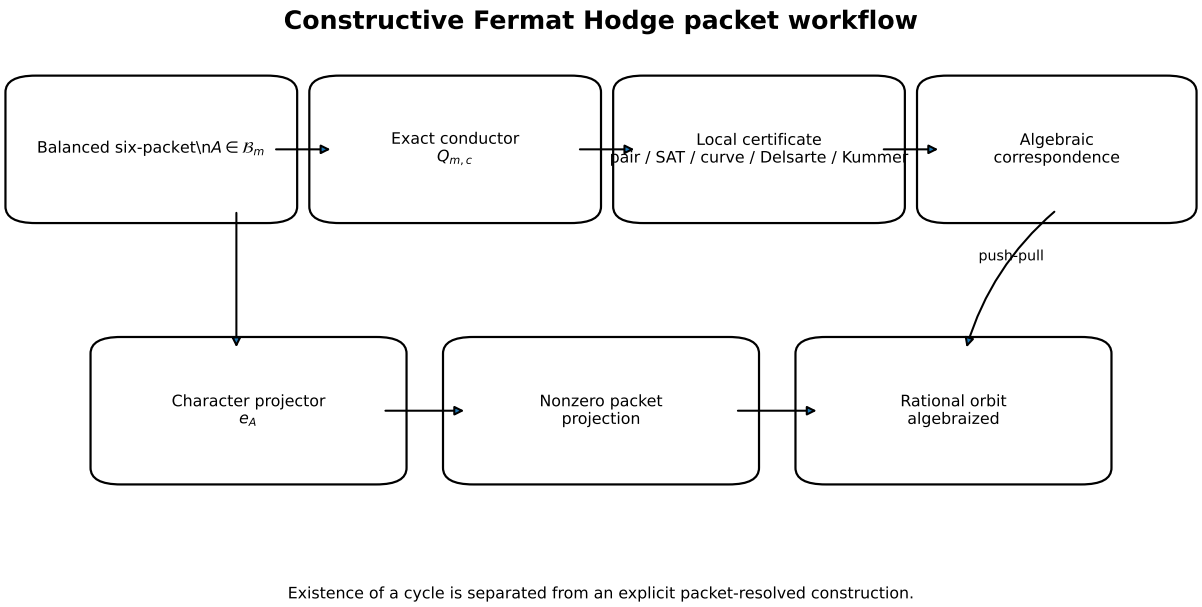


Figure 2.1: A constructive packet proof separates finite Hodge classification from the algebraic correspondence that realizes the packet.

2.1 Centered Fourier form

Define

$$\rho_m(r) = \langle r \rangle_m - \frac{m}{2}, \qquad F_a(u) = \sum_i \rho_m(ua_i).$$

Then a is balanced if and only if $F_a(u) = 0$ for all units. Moreover,

$$F_a(-u) = -F_a(u),$$

so only odd characters occur. For a Dirichlet character χ on U_m, put

$$\mathcal{B}_{m,\chi}(r) = \sum_{u \in U_m} \rho_m(ur)\overline{\chi(u)}.$$

Finite Fourier inversion gives the exact criterion

$$\boxed{a \in \mathcal{B}_m \iff \sum_i \mathcal{B}_{m,\chi}(a_i) = 0 \quad \text{for every odd } \chi.} \tag{BF}$$

This formulation is the basis for the prime-power classification and the conductor-defect tree.

Chapter 3

Exact conductor polyphase and divisor-lattice projectors

For a packet a, define

$$g(a) = \gcd(m, a_0, \dots, a_5), \qquad c(a) = \frac{m}{g(a)}.$$

Writing $a = g(a)b$, the coordinate-power map

$$\pi_{g(a)} : X_m^4 \longrightarrow X_{c(a)}^4, \qquad [x_i] \longmapsto [x_i^{g(a)}],$$

sends the degree-$c(a)$ character packet b to the degree-m packet a.

Theorem 3.1 (Conductor descent)**.** *Every imprimitive balanced packet at degree m is the coordinate-power pullback of a unique full-conductor packet at its exact conductor $c \mid m$. Consequently*

$$\mathcal{B}_m = \bigsqcup_{c|m} L_{m/c}\mathcal{B}_c^{\text{full}}.$$

Proof. If every occupied residue is divisible by g, divide the six entries and the modulus by g. Since

$$\langle ugb_i\rangle_{gc} = g\langle \bar{u}b_i\rangle_c,$$

the balance equation descends. The maximal common divisor is unique. The converse follows by scaling. □

For $d \mid m$, let

$$E_{m,d} = \frac{1}{(m/d)^5}\pi_{m/d}^*\pi_{m/d*}.$$

On a character of exact conductor c,

$$E_{m,d} = \begin{cases} 1, & c \mid d, \\ 0, & c \nmid d. \end{cases}$$

Therefore

$$\boxed{E_{m,d}E_{m,e} = E_{m,\gcd(d,e)}.} \tag{C1}$$

Möbius inversion produces pairwise orthogonal exact-conductor projectors

$$\boxed{Q_{m,c} = \sum_{d|c} \mu(c/d)E_{m,d}.} \tag{C2}$$

They satisfy

$$Q_{m,c}Q_{m,e} = \delta_{ce}Q_{m,c}, \qquad \sum_{c|m} Q_{m,c} = I.$$

Hence

$$\boxed{H^4_{\mathrm{prim}}(X^4_m, \mathbf{Q}) = \bigoplus_{c|m} \operatorname{im} Q_{m,c}.} \tag{8.3}$$

Each $Q_{m,c}$ is a rational linear combination of graph correspondences, so the conductor decomposition is itself algebraic.

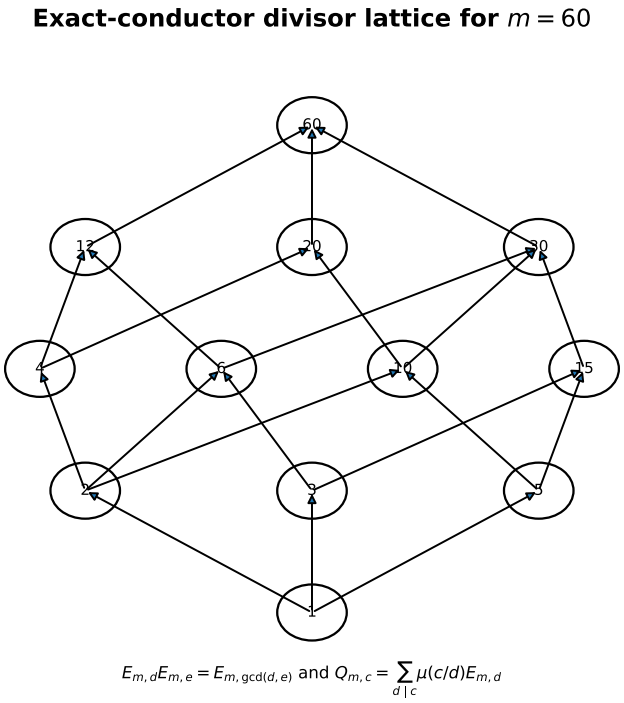


Figure 3.1: The divisor lattice is the correct all-degree progression. Only full-conductor seeds require new analysis.

Part II. Constructive Packet Grammar

Chapter 4

Constructive certificate grammar

Fix a full-conductor orbit in $\mathcal{B}_m$. The constructive program tests a hierarchy of increasingly general certificate channels.

4.1 Complementary pairs

For $1 \le r < m$, let

$$p_r = e_r + e_{m-r}.$$

A packet is pair-decomposable when

$$n = p_{r_1} + p_{r_2} + p_{r_3}. \tag{P}$$

These are the standard linear-cycle or Tate-pair packets.

4.2 Balanced four-packets and saturation

Let

$$\mathcal{F}_m = \left\{ \gamma \in \mathbf{N}^{m-1} : \mathbf{1}^T \gamma = 4, \quad R_m \gamma = 2m\,\mathbf{1} \right\}.$$

A saturated certificate is an exact nonnegative identity

$$\boxed{n + \sum_{j=1}^{2N-3} p_{r_j} = \sum_{i=1}^{N} \gamma_i, \qquad \gamma_i \in \mathcal{F}_m.} \tag{S}$$

The case $N = 2$ is one-step quasi-decomposition. Higher N allows several surface packets to reconstruct one six-packet after Tate stabilization.

Under the standard associative Fermat join formalism, (9.2) yields an isomorphism of character motives after tensoring with Tate factors. Since the right side is built from algebraic classes on Fermat surfaces, Tate cancellation returns algebraicity of the original packet. The exact Chow-level associativity and cancellation maps should be written explicitly in a final formal treatment.

4.3 Dual charges

An antisymmetric charge is a rational function w on residues satisfying

$$w(m-r) = -w(r)$$

and

$$\sum_{r\in\gamma} w(r) = 0 \qquad \forall \gamma \in \mathcal{F}_m.$$

Such a charge vanishes on every complementary pair and on every saturated identity. If $w(n) \neq 0$, the entire lower-packet saturation semigroup is excluded. This is a universal obstruction to a grammar, not a nonalgebraicity theorem.

4.4 Curve, Delsarte, and incidence channels

A curve certificate splits the six entries into triples

$$a = A \sqcup B, \qquad \sum A = m, \quad \sum B = 2m, \quad B = uA \pmod{m}.$$

The resulting $(1,1)$ class on a product of Fermat curves is algebraic by Lefschetz $(1,1)$, and the join correspondence returns a surface class.

A Delsarte certificate consists of an exponent matrix and a resolved monomial map to a lower-degree hypersurface carrying an already algebraic packet. An incidence certificate uses algebraic lowering maps

$$A_r : V \to H^2(Y_r)$$

whose combined positive transfer

$$T = \sum_r A_r^\dagger A_r$$

is invertible on the packet. Divisor lowering and Cayley–Hamilton reconstruction then produce the cycle.

Chapter 5

Positive packet frames and exact synthesis

5.1 Analysis and synthesis

Fix a finite family of rational algebraic cycle classes

$$c_1, \dots, c_N \in H^{2p}(X, \mathbf{Q}).$$

Let $V \subseteq H^{2p}(X, \mathbf{R})$ be the target packet under study. Define

$$C : \mathbf{R}^N \to H^{2p}(X, \mathbf{R}), \qquad Ce_j = c_j,$$

and the positive frame operator

$$S = CC^* = \sum_{j=1}^{N} |c_j\rangle\langle c_j|.$$

Theorem 5.1 (Finite positive cycle-frame certificate)**.** *Let $A = \operatorname{span}_{\mathbf{R}}\{c_1, \dots, c_N\}$. Then*

$$\operatorname{im} S = A, \qquad \ker S = A^{\perp}.$$

The selected cycles span V if and only if $S|_V$ is positive definite.

Proof. The equality $\operatorname{im}(CC^*) = \operatorname{im} C$ is standard. Moreover

$$\langle v, Sv\rangle = \|C^*v\|^2 = \sum_j |\langle c_j, v\rangle|^2.$$

This vanishes exactly when v is orthogonal to every cycle class. □

If H is a matrix whose columns form an orthonormal basis of V, the finite Gram certificate is

$$G = H^*SH.$$

Then $A \supseteq V$ if and only if $\lambda_{\min}(G) > 0$.

The orthogonal synthesis of a class α from the cycle span is

$$P_A\alpha = SS^{\dagger}\alpha.$$

If $\alpha \in A$, this is exact.

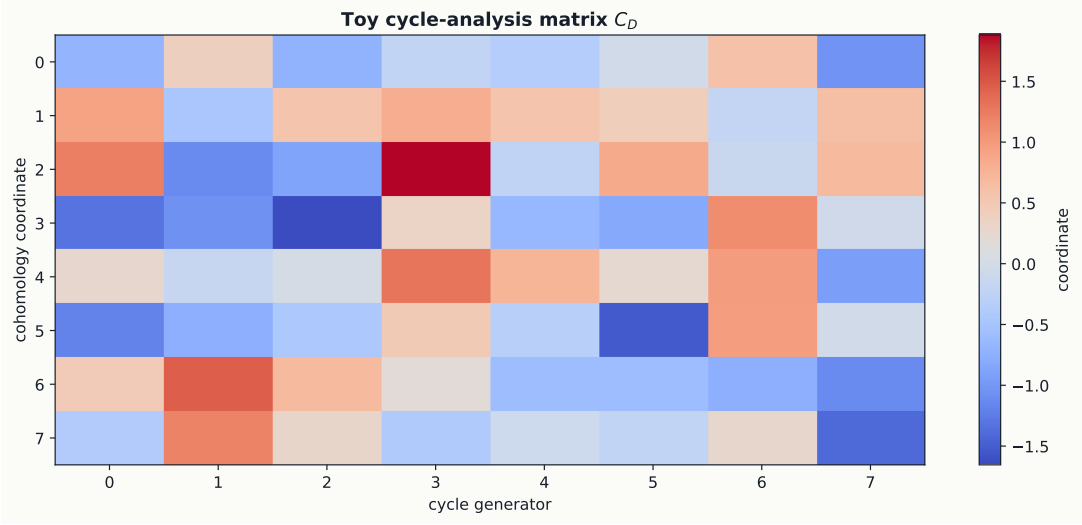


Figure 5.1: A finite cycle-analysis matrix records cohomology coordinates of candidate algebraic cycles.

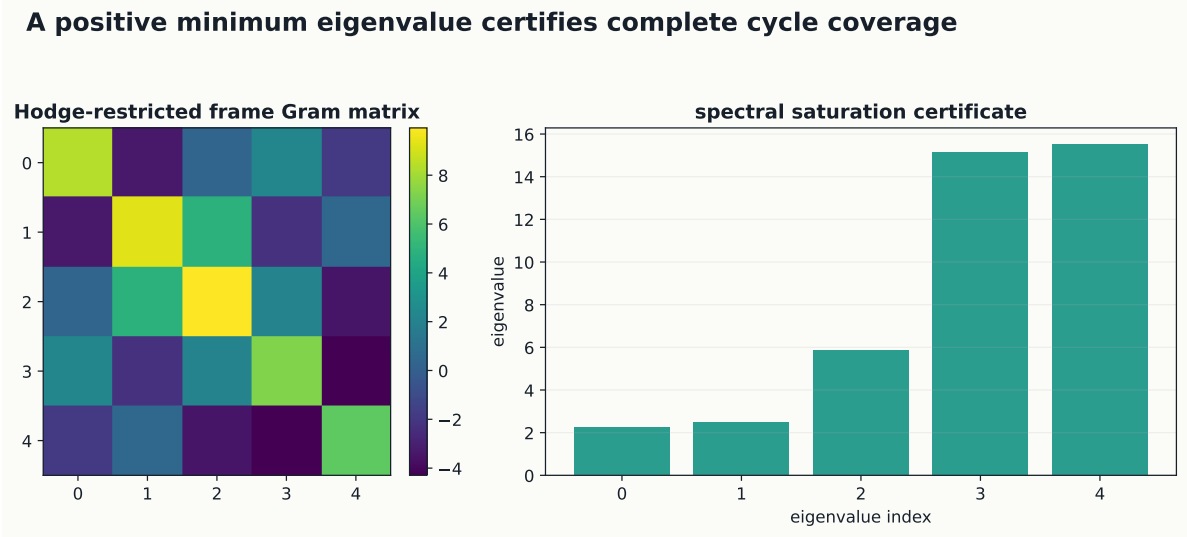


Figure 5.2: The packet-restricted Gram spectrum gives a finite spectral saturation certificate.

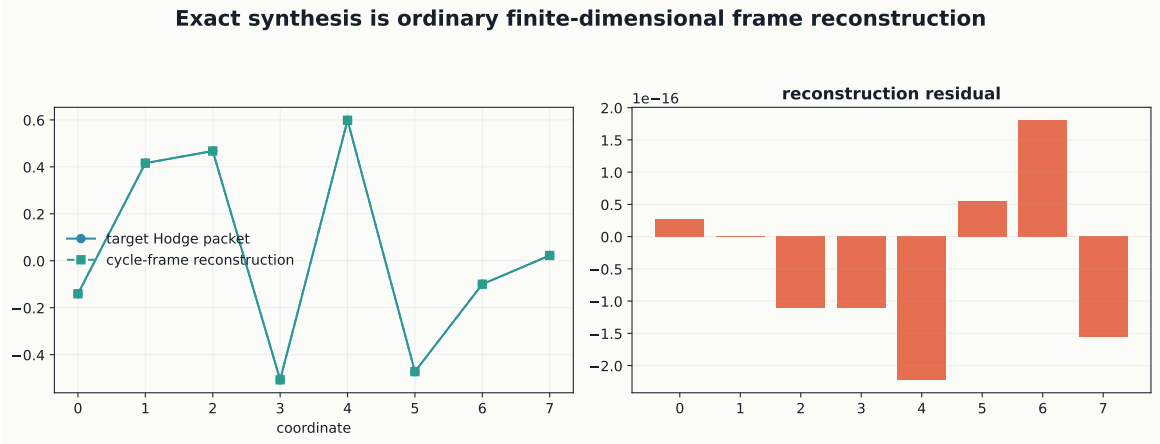


Figure 5.3: Finite-dimensional frame synthesis reconstructs the class whenever it lies in the algebraic span.

5.2 Packet assembly without false orthogonality

Partition the cycle indices into packets I_λ. Define

$$S_\lambda = \sum_{j \in I_\lambda} |c_j\rangle\langle c_j|.$$

Then

$$S = \sum_\lambda S_\lambda.$$

No orthogonality of the child spans is assumed.

Theorem 5.2 (Positive child-kernel intersection)**.** *For any subspace V,*

$$\ker(S|_V) = \bigcap_\lambda \ker(S_\lambda|_V).$$

Proof. If $Sv = 0$, positivity gives

$$0 = \langle v, Sv\rangle = \sum_\lambda \langle v, S_\lambda v\rangle.$$

Each nonnegative summand vanishes, which implies $S_\lambda v = 0$. The converse is immediate. □

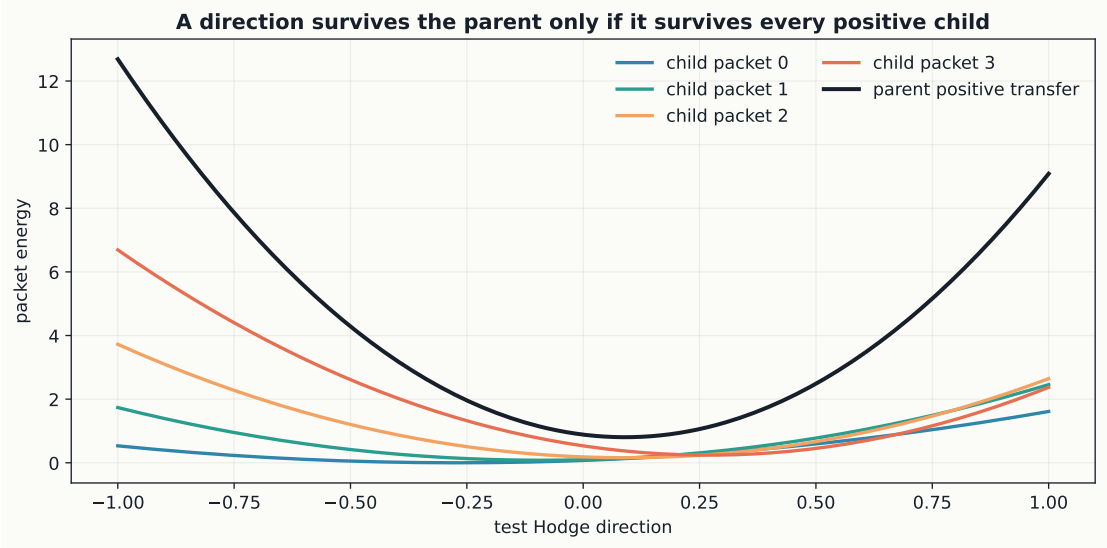


Figure 5.4: A target direction survives the parent obstruction only when every positive child channel misses it.

Chapter 6

Polyphase refinement of cycle and correspondence libraries

6.1 Exact parent-child frame law

Fix a projective embedding and assign each cycle its degree. For modulus M, define the residue packet

$$I_{M,r} = \{j : \deg Z_j \equiv r \pmod{M}\}, \qquad S_{M,r} = \sum_{j \in I_{M,r}} |c_j\rangle\langle c_j|.$$

Theorem 6.1 (Exact degree polyphase refinement)**.** *For every positive integer L,*

$$S_{M,r} = \sum_{\ell=0}^{L-1} S_{LM,r+\ell M}.$$

Proof. The congruence class r modulo M is the disjoint union of the classes $r + \ell M$ modulo LM. The rank-one frame terms are partitioned accordingly. □

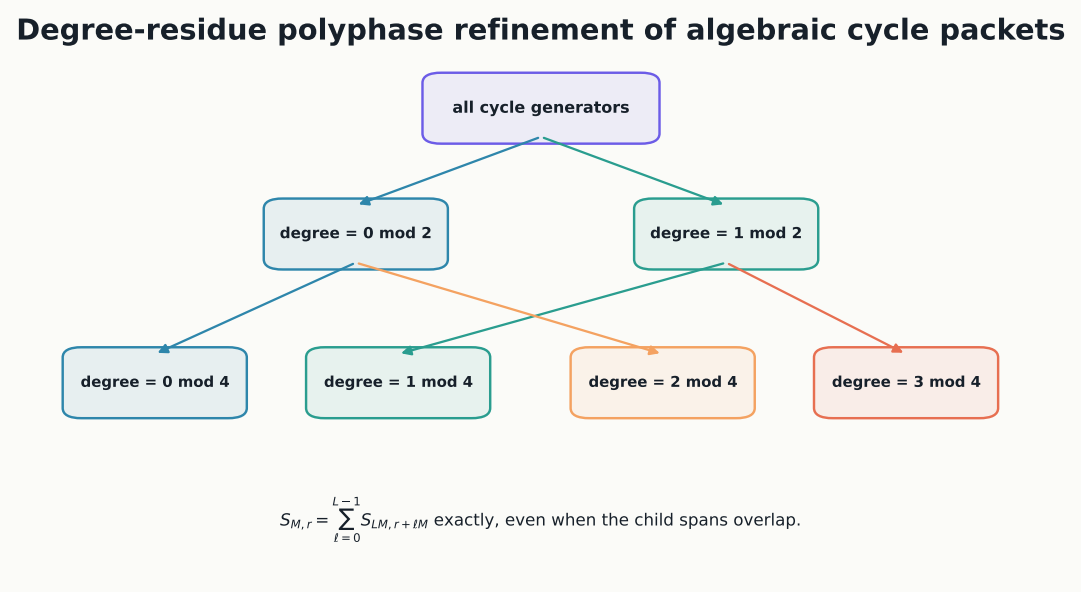


Figure 6.1: Degree addresses refine from modulo 2 to modulo 4 while the total cycle frame remains unchanged.

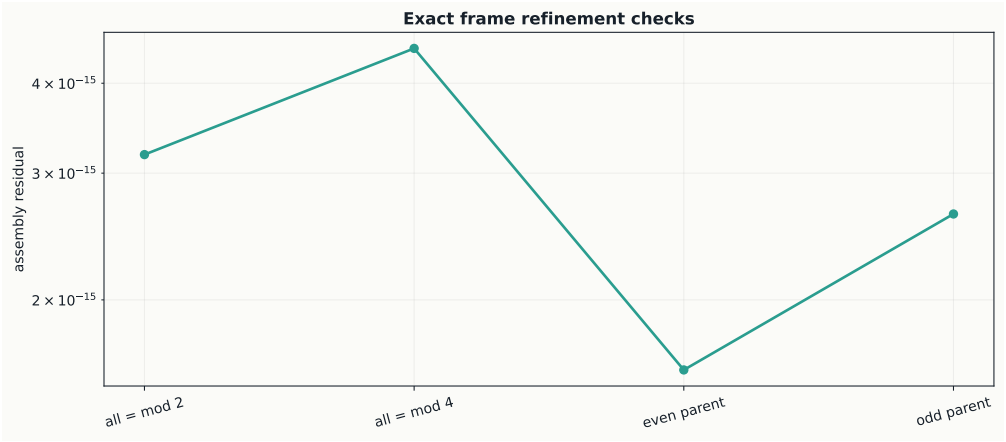


Figure 6.2: Numerical certificates of exact parent-child frame assembly in the reproducible toy model.

6.2 Multiaxial packet addresses

Degree residue is only one address. A refined cycle packet may be labeled by

$$\lambda = (r_{\mathrm{Lef}}, r_{\deg}, \chi_{\mathrm{mon}}, s_{\mathrm{supp}}, \nu_{\mathrm{Frob}}).$$

The entries record Lefschetz depth, degree residue, monodromy or symmetry type, support stratum, and arithmetic phase. The total positive frame remains the sum of all leaves. This allows independent methods to solve different geometric sectors before exact synthesis.

Chapter 7

Finite saturation of constructive cycle libraries

For each degree cutoff D, let $A^p_D(X)$ be the rational span of codimension-p cycle classes of degree at most D. Then

$$A^p_1 \subseteq A^p_2 \subseteq \cdots \subseteq H^{2p}(X, \mathbf{Q}).$$

Theorem 7.1 (Finite stabilization)**.** *There exists D_0 such that $A^p_D(X) = A^p_{D_0}(X)$ for every $D \geq D_0$.*

Proof. The dimensions form a nondecreasing sequence of integers bounded by $\dim H^{2p}(X, \mathbf{Q})$. Only finitely many strict increases are possible. □

Corollary 7.2 (Approximation becomes exact)**.** *Let P_D be the orthogonal projector onto $A^p_D(X) \otimes \mathbf{R}$. For a rational class α,*

$$\lim_{D \to \infty} \|(I - P_D)\alpha\| = 0$$

if and only if $\alpha \in A^p_D(X)$ for some finite D.

Proof. After stabilization, the residual is constant. A constant tail converging to zero is zero. □

Because the matrices and target vector can be chosen over $\mathbf{Q}$, real membership implies rational membership by row reduction over $\mathbf{Q}$.

Finite packet-coverage certificate

If H is a rational basis matrix for a selected target packet and C_D is the cycle matrix, then the degree-D cycle library covers that packet exactly when

$$\operatorname{rank}_{\mathbf{Q}} C_D = \operatorname{rank}_{\mathbf{Q}} [C_D \ H].$$

Equivalently, the packet-restricted positive frame has positive minimum eigenvalue.

Part III. Exceptional Transfers, Prime-Local Structure, and Infinite Families

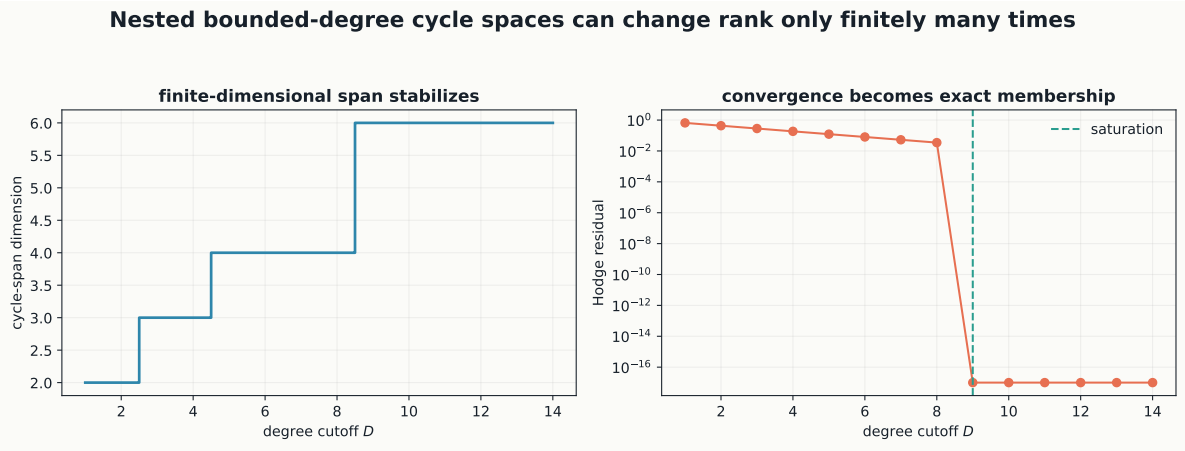


Figure 7.1: The cycle span has a finite rank ceiling. Once the residual converges to zero, exact membership has already occurred at a finite stage.

Chapter 8

The exceptional degree-33 packet and the Klein–Delsarte transfer

The degree-33 Fermat fourfold provides a stringent constructive test because one rational orbit escapes the ordinary decomposable and standard channels. The packet is first isolated and separated from several obvious cycle families. A cross-family monomial transfer to a smooth cubic fourfold then supplies the missing correspondence channel.

8.1 The exceptional packet

A representative is

$$a = (7, 10, 13, 19, 22, 28),$$

or after permutation,

$$\widetilde{a} = (7, 19, 28, 10, 13, 22).$$

Every unit conjugate has residue sum $99 = 3 \cdot 33$, so the orbit is rational of type $(2, 2)$. It is not totally pair-decomposable. The previously proposed symmetric surface

$$W = \{p_1(x^{11}) = p_2(x^{11}) = p_3(x^{11}) = 0\}$$

does not carry the primitive packet: Newton's identity makes the third equation redundant on X^4_{33}, and

$$[W] = 242h^2.$$

Thus W is purely ambient.

8.2 Klein–Delsarte cubic

Consider the smooth cubic fourfold

$$Y: \quad y_0^2y_4 + y_1^2y_0 + y_2^2y_1 + y_3^2y_2 + y_4^2y_3 + y_5^3 = 0.$$

An explicit monomial exponent matrix gives a dominant rational map

$$\phi : X^4_{33} \dashrightarrow Y$$

whose projective-torus lattice map has Smith diagonal

$$(1, 33, 33, 33, 33),$$

so the generic torus degree is 33^4.

Take the cubic Jacobian-ring class

$$P(y) = y_1 y_2 y_5.$$

It is nonzero in the degree-three Jacobian ring. The residue pullback has the exact exponent identity

$$C^T b + 10\mathbf{1} = (6, 18, 27, 9, 12, 21) = \widetilde{a} - \mathbf{1}. \tag{10.1}$$

Therefore the pullback of the corresponding primitive cubic $(2, 2)$-class lies in the one-dimensional Fermat line $V(\widetilde{a})$.

Because the graph is defined over $\mathbf{Q}$, its image is Galois-stable. Containing one character line forces it to contain the complete rational orbit. Rational $(2, 2)$ classes on the cubic are algebraized through its Fano variety of lines, and the graph correspondence transports those cycles back to the Fermat fourfold.

Theorem 8.1 (Degree-33 packet closure)**.** *The rational packet generated by* $(7, 10, 13, 19, 22, 28)$ *is algebraic on* X^4_{33}.

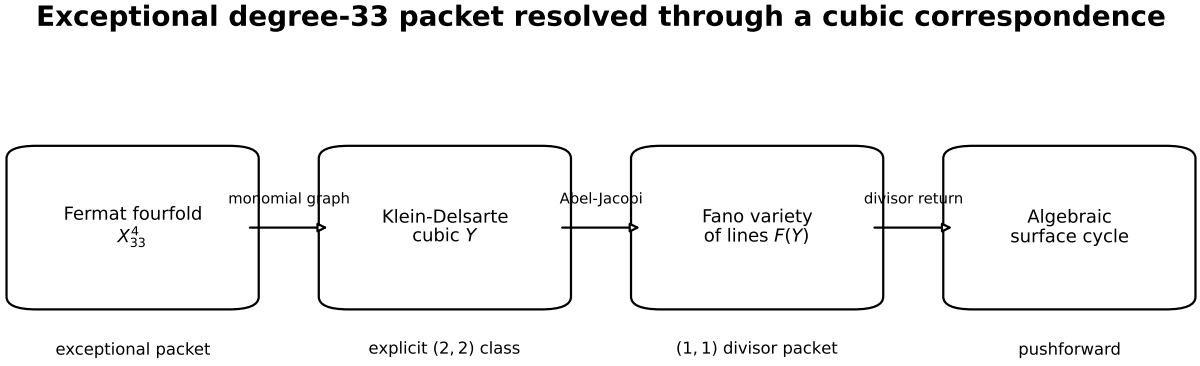


Figure 8.1: The exceptional degree-33 packet is reached through a lower-degree cubic and a divisor-lowering incidence correspondence.

The episode is methodologically important. Failure of a finite cycle grammar is not evidence of nonalgebraicity. It may indicate a missing cross-family correspondence.

Chapter 9

The cubic-fourfold transfer engine

Let $X \subset \mathbb{P}^5$ be a smooth cubic fourfold and let $F(X)$ be its Fano variety of lines. The universal incidence variety

$$P = \{(\ell, x) : x \in \ell\} \subset F(X) \times X$$

induces the Abel-Jacobi map

$$A = p_* q^* : H^4_{\mathrm{prim}}(X, \mathbf{Q}) \to H^2_{\mathrm{prim}}(F(X), \mathbf{Q}).$$

Beauville and Donagi proved that this is a polarized Hodge isomorphism up to the standard twist [2]. The polarized adjoint is represented by an algebraic incidence return after inserting the appropriate power of the Pluecker class. Therefore the divisor-lowering transfer is invertible.

Classical validation

A primitive rational $(2,2)$ class on a cubic fourfold is sent to a rational $(1,1)$ class on $F(X)$, represented by a divisor. The algebraic return correspondence and its polynomial inverse reconstruct an algebraic codimension-two class on X. This is a validation of the packet criterion, not a claim of a new proof of the classical cubic-fourfold result.

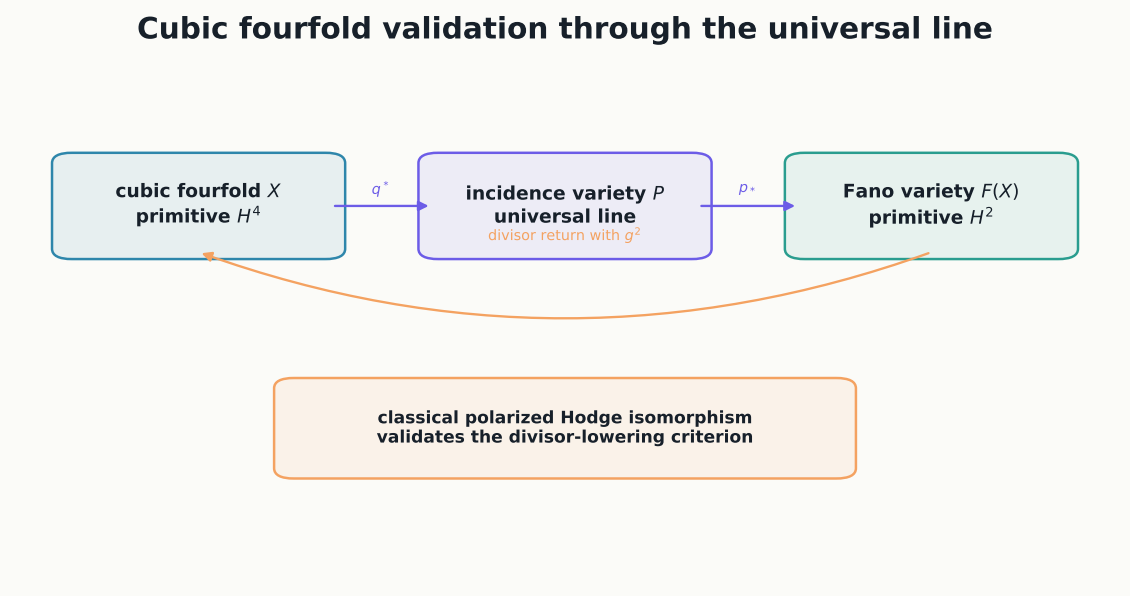


Figure 9.1: The universal line lowers primitive degree-four cohomology to divisor degree on the Fano variety.

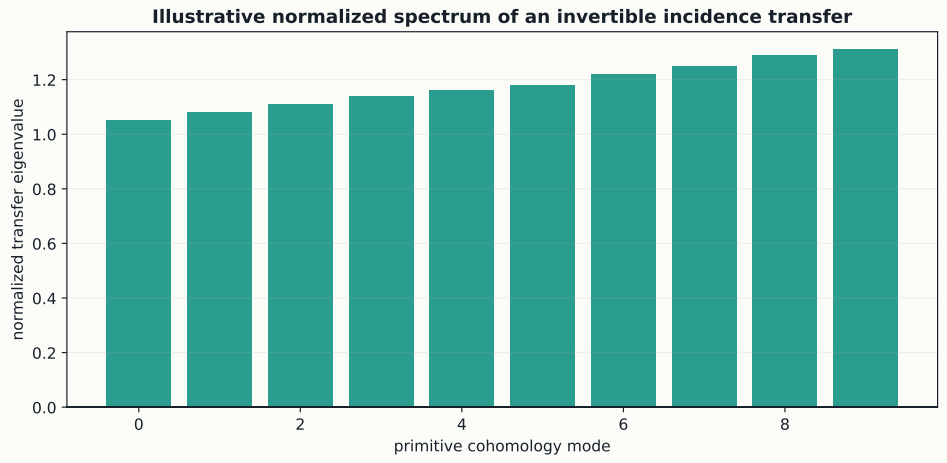


Figure 9.2: Illustrative normalized transfer spectrum. The mathematical criterion is nonvanishing, not the displayed toy values.

Chapter 10

Infinite towers and full-conductor promotion

The coordinate-power maps propagate every solved seed through infinitely many degrees.

Theorem 10.1 (Infinite packet tower)**.** *Let W be an algebraic rational character packet on X_d^{2p}. For every $k \geq 1$, its coordinate-power pullback is an algebraic packet on X_{kd}^{2p}. In particular, the degree-33 exceptional packet yields a 20-dimensional algebraic packet on every X_{33k}^4.*

The tower packet remains an old conductor-33 sector. A stronger construction appends a complementary pair

$$(1, m-1)$$

after scaling a solved seed to modulus $m = kd$. If

$$b = (b_0, \ldots, b_{2p+1})$$

is algebraic at degree d, then

$$\Theta_{m,b} = (kb_0, \ldots, kb_{2p+1}, 1, m-1)$$

has full raw conductor m and middle Hodge type two dimensions higher. Its rational orbit has dimension $\varphi(m)$. This produces full-raw-conductor algebraic packets of unbounded rank.

Raw conductor is not the complete novelty invariant: removing Tate pairs or reversing joins can reveal a smaller stable conductor. The stable residual should therefore be defined only after eliminating conductor descent, Tate suspension, joins, Delsarte pullbacks, and incidence channels.

Chapter 11

Prime-power classification

The odd prime-power classification follows from a primitive-character filtration and a six-entry mass bound.

11.1 Valuation strata and orbit mass

Let $m = p^r$ with p odd. Decompose the entries by exact p-adic valuation. At one stratum, write $a = p^j b$, with $b \in U_{p^{r-j}}$, and define the odd count

$$\delta_j(b) = n_j(b) - n_j(-b).$$

Since the packet has six entries,

$$\|\delta_j\|_1 \le 12. \tag{12.1}$$

Primitive odd characters force δ_j to be constant on the kernel cosets of

$$U_{p^s} \to U_{p^{s-1}}.$$

A nonzero odd kernel-orbit defect has mass at least $2p$.

11.2 Large odd primes

If $p \ge 7$, then $2p > 12$, so every odd defect vanishes. Each residue occurs with the same multiplicity as its complement. Therefore every packet is three complementary pairs.

11.3 The three-primary branch

At $p = 3$, a kernel orbit has three elements. Define

$$O_b = \{b, b+q, b+2q\}, \qquad q = 3^{r-1},$$

and

$$\Gamma_b = O_b \sqcup \{3(q-b)\}.$$

The packet Γ_b is a balanced four-packet. A case analysis on the number of oriented kernel orbits gives exact identities expressing every full-conductor nonpair packet as a pair/surface packet or a one-step saturated sum of two balanced four-packets.

11.4 The five-primary branch

At $p = 5$, a nonzero odd defect consumes exactly five unit entries. The sixth entry is forced by balance. Up to the unit action there is one full-conductor exceptional orbit

$$\boxed{\alpha_{5^r} = (1, 1+q, 1+2q, 1+3q, 1+4q, 5q-5), \qquad q = 5^{r-1}.} \tag{12.2}$$

All other packets lie in the pair/surface grammar. The family (12.2) is the pure five-primary Hasse–Davenport branch.

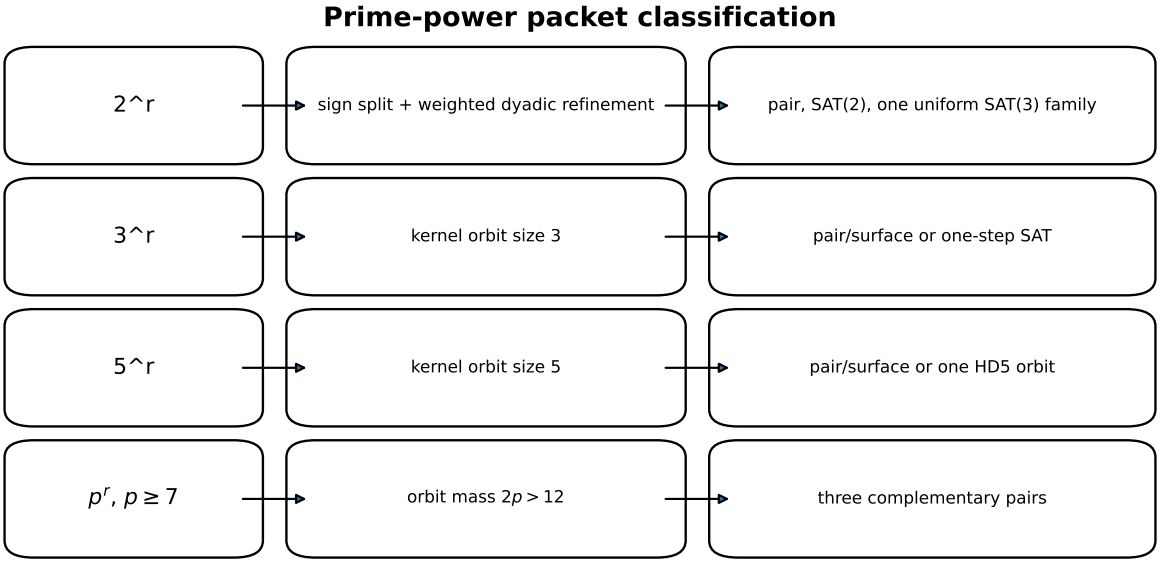


Figure 11.1: The six-entry mass bound produces a complete odd prime-power classification and isolates the five-primary family.

m	Full orbits	Pair/SAT(2)	Residual
49	124	124	0
81	436	436	0
125	827	826 pair	1

Chapter 12

Weighted dyadic reconstruction

Prime 2 cannot be handled by inserting $p = 2$ into the odd-prime orbit-mass proof. The unit group has the separate form

$$\mathbf{Z}_2^\times \cong \{\pm 1\} \times (1 + 4\mathbf{Z}_2).$$

The first address is a sign packet; only afterward does the principal-unit refinement begin.

Let $m = 2^s$ and $h = 2^{s-1}$. For an odd residue $b \pmod h$, its two lifts are b and $b + h$. Define

$$L_s n(b) = n(b) + n(b + h), \qquad D_s n(b) = n(b) - n(b + h).$$

For every odd unit u, one has the exact weighted identity

$$\boxed{\begin{aligned} &n(b)\rho_s(ub) + n(b+h)\rho_s(u(b+h)) \\ &\qquad = L_s n(b)\rho_{s-1}(\bar{u}b) + 2^{s-2} D_s n(b)\varepsilon_s(u, b), \end{aligned}} \tag{13.1}$$

where $\varepsilon_s(u, b) \in \{\pm 1\}$. Repetition yields a Haar-like expansion whose detail packets carry visible powers of two. The Hodge balance equation becomes a binary carry relation between weighted details.

Exact enumeration gives:

m	Full orbits	Pair	SAT(2)	SAT(3)	Unresolved
8	13	9	4	0	0
16	45	27	18	0	0
32	134	91	42	1	0
64	419	331	87	1	0
128	1435	1259	175	1	0

For $m = 32, 64, 128$, the unique packet left after pair and SAT(2) reduction is

$$A_m = \left(1, \frac{m}{4} + 1, \frac{m}{2} + 1, \frac{m}{2} + 4, \frac{3m}{4} + 1, m - 8\right).$$

It obeys the uniform SAT(3) identity

$$\begin{aligned} &A_m + (2, m-2) + (4, m-4) + (m/2, m/2) \\ &= (1, m/2, m/2+1, m-2) \\ &\quad + (2, m/4+1, 3m/4+1, m-4) \\ &\quad + (4, m/2, m/2+4, m-8). \end{aligned} \tag{13.2}$$

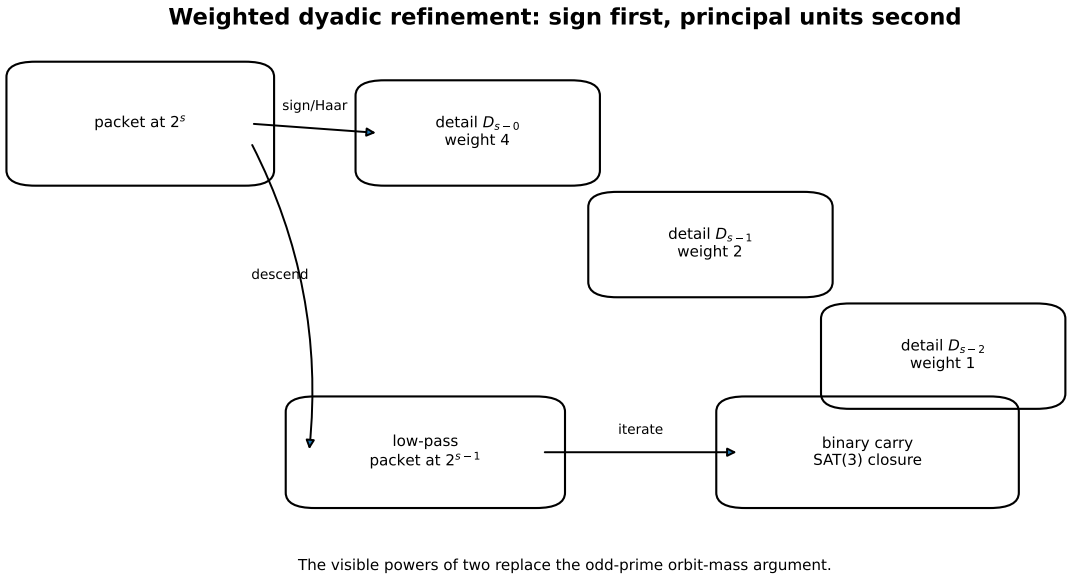


Figure 12.1: At prime two, sign refinement and powers-of-two weights replace the odd-prime orbit-size argument.

All three packets on the right are balanced. Formula (13.2) was checked through modulus 4096. A uniform proof that no second residual family appears at higher dyadic depth remains an important formalization target, although the computational evidence is strong.

Chapter 13

Finite constructive audit through degree 100

The finite constructive audit exactly enumerates balanced packets, reduces them by unit orbits, checks pair and SAT(2) certificates, and searches higher saturated identities with $N \leq 8$. Every returned identity is rechecked by exact integer arithmetic.

All named verification groups pass:

Suite	Passed	Total
projector resolved hodge calculus	18	18
fermat33 klein closure	14	14
infinite klein tower	4	4
conductor polyphase	4	4
odd prime power audit	13	13
Total	53	53

The dyadic weighted identity passes 89,478,480 coefficient checks through 2^{14}. The exact packet audit is complete through degree 100 for the implemented grammar.

After removing the degree-33 Klein channel and the degree-39 curve channel, every remaining post-saturation degree in the audit is divisible by five. This is a powerful localization of the residual grammar, but it is not by itself a universal proof.

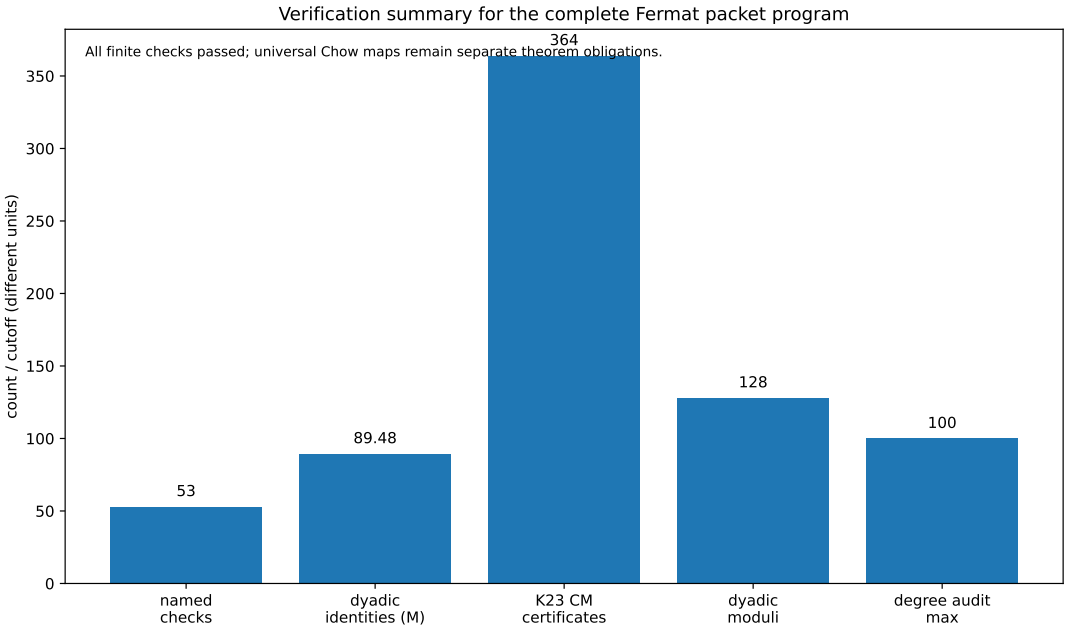


Figure 13.1: Finite checks validate the symbolic grammar and examples; universal Chow correspondences remain separate theorem obligations.

13.1 Exact residual representatives

Degree	Post-saturation representatives
25	(1,6,11,16,20,21)
33	(1,4,16,22,25,31)
35	(1,8,15,22,29,30)
39	(1,7,16,22,34,37)
40	(1,9,17,25,33,35)
45	(1,10,19,28,37,40); (1,19,20,28,30,37)
50	(1,7,27,30,41,44); (1,11,21,31,41,45); (1,13,22,32,40,42)
55	(1,12,23,34,45,50)
65	(1,14,27,40,53,60)
70	(1,6,43,45,57,58); (1,15,29,43,57,65); (1,20,24,42,61,62); (1,20,29,43,57,60); (1,29,35,43,45,57)
75	(1,16,31,46,61,70)
80	(1,17,33,49,65,75)
85	(1,18,35,52,69,80)
90	(1,8,26,62,85,88); (1,19,37,55,73,85); (1,19,37,60,73,80); (1,19,37,65,73,75); (1,23,38,55,74,79); (1,38,46,55,56,74); (2,38,45,55,56,74)
95	(1,20,39,58,77,90)
100	(1,21,41,61,81,95)

The residuals are not random. Their mod-5 phase profiles form a small local grammar.

Part IV. CRT–Kummer Polyphase and CM Half-Twists

Chapter 14

Local mod-5 phase grammar

Assume $m = 5q$ and first suppose $(5, q) = 1$. For $b \pmod q$ and a phase $r \in \{0, 1, 2, 3, 4\}$, let $\lambda_r(b)$ be the CRT lift satisfying

$$\lambda_r(b) \equiv r \pmod 5, \qquad \lambda_r(b) \equiv b \pmod q.$$

The finite audit reveals four canonical multiplicity profiles, up to multiplication by a unit modulo five:

$$(1, 0, 0, 0, 5), \quad (2, 1, 1, 1, 1), \quad (1, 0, 2, 1, 2), \quad (1, 0, 1, 3, 1).$$

The first is the complete-five-phase K5 packet, the second is a four-phase K4 lift, and the final two are one K23 pair–triple mechanism.

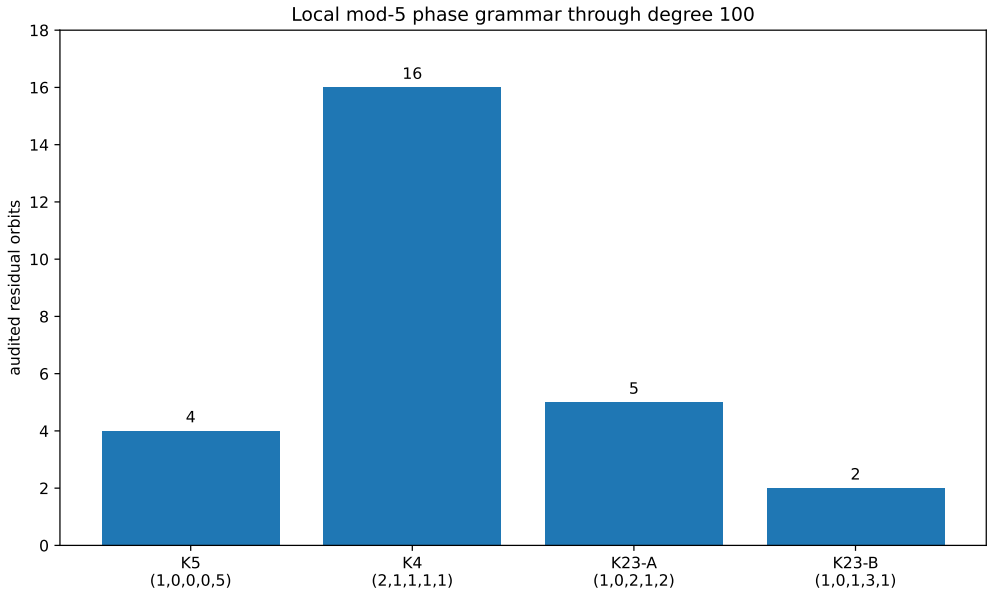


Figure 14.1: Only four local mod-5 profiles occur among the audited residual orbits; the last two unify as K23.

14.1 K4 four-phase reduction

Choose nonzero residues $b, c, d \pmod q$ and put

$$e \equiv -5^{-1}b \pmod q.$$

Define

$$A_{b;c,d} = (\lambda_1(b), \lambda_2(b), \lambda_3(b), \lambda_4(b), 5c, 5d)$$

and

$$\Gamma_{b;c,d} = (b, c, d, e).$$

Theorem 14.1 (Four-phase CRT reduction)**.** *The six-packet $A_{b;c,d}$ is balanced modulo $5q$ if and only if the four-packet $\Gamma_{b;c,d}$ is balanced modulo q.*

Proof. For a unit u, let $s = \langle ub \rangle_q$ and $z = \langle u5^{-1}b \rangle_q$. The complete five-element fiber over s has sum $5s + 10q$; its phase-zero lift is $5z$. The four nonzero phases therefore sum to $5s + 10q - 5z$. Adding the last two entries and imposing the middle value $15q$ gives

$$s + \langle uc \rangle_q + \langle ud \rangle_q - z = q.$$

Since $\langle ue \rangle_q = q - z$, this is exactly the balance equation for (b, c, d, e). □

All 16 audited K4 residuals reduce exactly to balanced four-packets.

Chapter 15

K23 pair–triple factorization

Let a K23 packet have one phase-zero entry and five nonzero phases:

$$A = P \sqcup T \sqcup \{5e\}, \qquad |P| = 2, \quad |T| = 3,$$

with

$$\sum P \equiv 0 \pmod 5, \qquad \sum T \equiv 0 \pmod 5.$$

Write

$$\sum P = 5C, \qquad \sum T = 5D,$$

and reduce C, D modulo q to positive representatives c, d. Define

$$C_P = P \sqcup \{m - 5c\},$$

$$S_T = T \sqcup \{m - 5d\},$$

and

$$L = (5c, 5d, 5e).$$

Theorem 15.1 (Exact K23 packet identity)**.** *One has the exact multiset equality*

$$\boxed{A + (5c, m - 5c) + (5d, m - 5d) = C_P + S_T + L.} \tag{15.1}$$

For every unit $u \pmod m$*,*

$$h_{C_P}(u) + h_{S_T}(u) + h_L(u) = 5,$$

and $c + d + e \equiv 0 \pmod q$*.*

The five nonzero phases have only two irreducible possibilities: all five phases are equal, or they contain a complementary pair and a zero-sum triple. Indeed, if no complementary phase pair occurs, the phase support lies in one of four two-element subsets; the zero-sum condition forces all five entries to use one phase. This proves the local K5/K23 dichotomy.

At the motive level, write

$$N = M(C_P) \otimes M(S_T).$$

The Hodge-level identity implies that $N(1)$ has weight one and has Hodge type opposite to the lower curve packet $M(L)$ at every embedding.

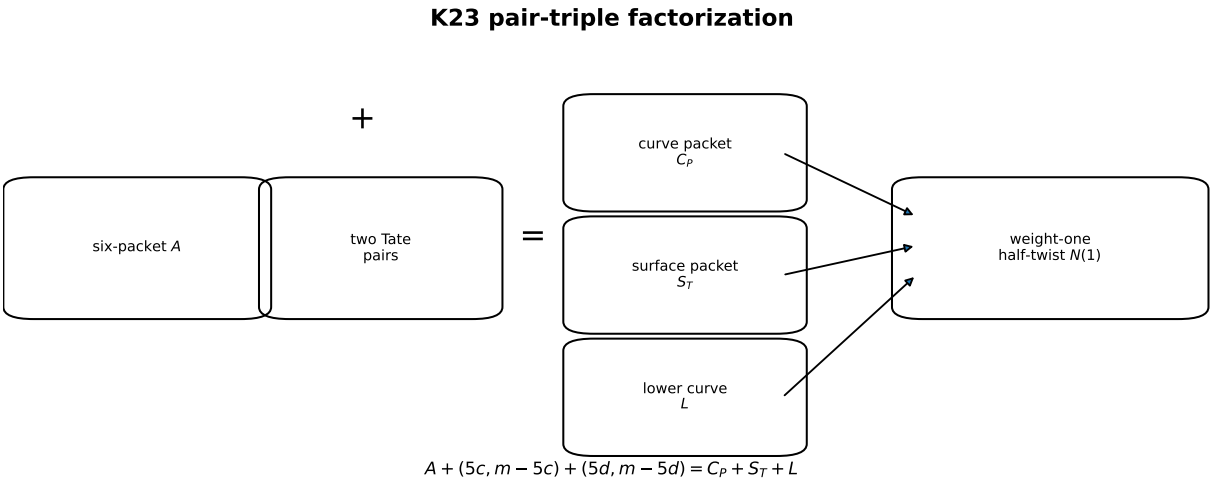


Figure 15.1: The K23 packet becomes a curve packet, a surface packet, and a lower curve packet after two explicit Tate-pair stabilizations.

Chapter 16

Imaginary CM selector

Write the lower curve packet as

$$\ell = (c, d, e) \pmod q, \qquad c + d + e \equiv 0 \pmod q.$$

Let $K = \mathbf{Q}(\zeta_q)$ and define

$$\boxed{\Delta_\ell = (1 - \zeta_q^c)(1 - \zeta_q^d)(1 - \zeta_q^e).} \tag{CM1}$$

Theorem 16.1 (Pure imaginary selector)**.**

$$\overline{\Delta_\ell} = -\Delta_\ell.$$

Proof. Since

$$1 - \zeta_q^{-a} = -\zeta_q^{-a}(1 - \zeta_q^a),$$

complex conjugation multiplies the product by

$$(-1)^3 \zeta_q^{-(c+d+e)} = -1.$$

□

For an embedding $\sigma_u : \zeta_q \mapsto \zeta_q^u$, let

$$h_\ell(u) = \frac{\langle uc \rangle_q + \langle ud \rangle_q + \langle ue \rangle_q}{q} \in \{1, 2\}.$$

Using

$$1 - e^{2\pi i r/q} = -2i\, e^{\pi i r/q} \sin \frac{\pi r}{q},$$

one obtains

$$\sigma_u(\Delta_\ell) = 8i(-1)^{h_\ell(u)} \prod_{a \in \{c,d,e\}} \sin \frac{\pi \langle ua \rangle_q}{q}. \tag{CM2}$$

Every sine factor is positive, so the sign of the imaginary part is exactly the Hodge sign. Consequently the CM type of $N(1)$ is the conjugate of the CM type of the lower curve packet:

$$\boxed{\Sigma_N = \overline{\Sigma_\ell}.} \tag{CM3}$$

This is the rigorous form of the statement that the half-twist occurs on the imaginary plane.

The exact audit generates 364 K23 certificates from seven rational packet orbits. Every certificate obeys the opposite curve CM sign, factors through a lower conductor, and is a Galois translate of an upper-half-plane CM type.

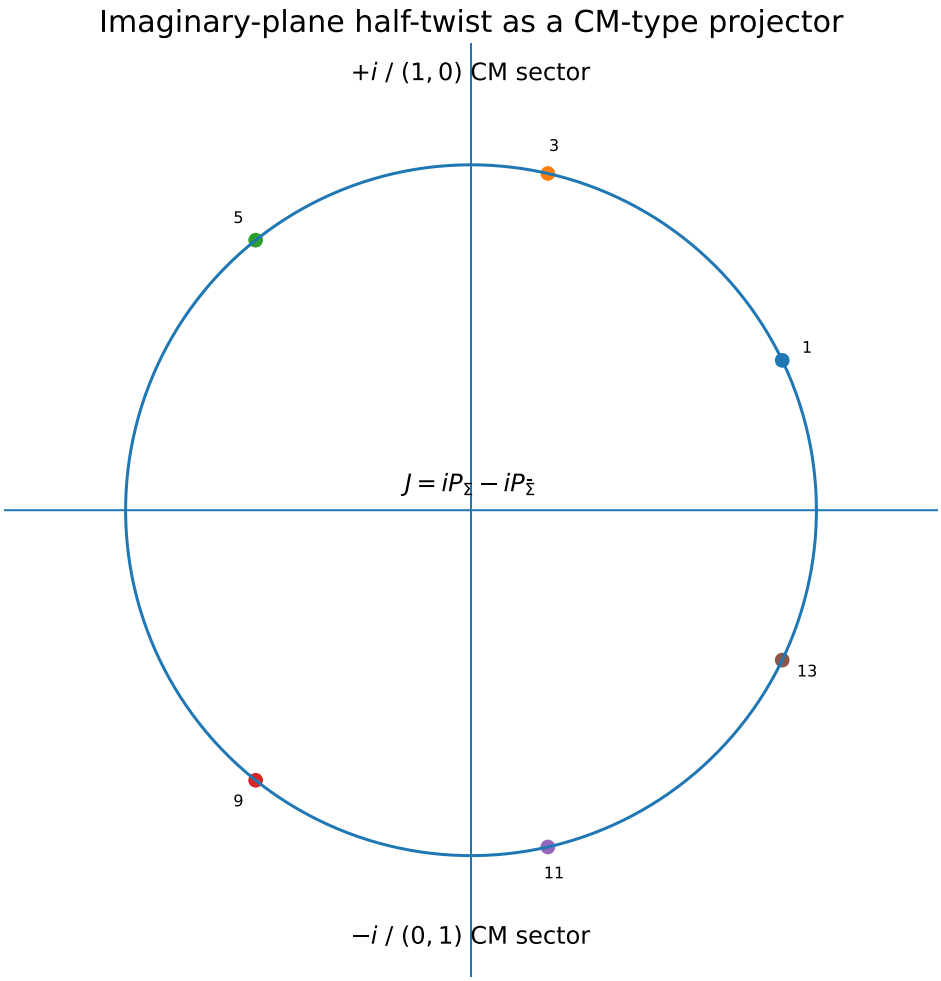


Figure 16.1: The half-twist acts with eigenvalues $+i$ and $-i$ on conjugate CM embedding sectors.

Item	Value
K23 rational orbits	7
Exact K23 certificates	364
Opposite curve CM sign	True
Lower-conductor factorization	True
Upper-half-plane CM translate	True

In the audited cases, the minimal CM conductors are $3, 5, 7$, or 9, even when the original fourfold modulus is $50, 70$, or 90. Thus the weight-one carrier is not arbitrary: it is a character factor of the Jacobian of a lower-degree Fermat curve.

Chapter 17

Explicit Fermat-curve carrier and the rank-one Chow target

Let F_d be the Fermat curve of the minimal CM conductor and let e_{Σ_N} be the algebraic character idempotent selecting the required orbit. The explicit carrier is

$$A_{\Sigma_N} = e_{\Sigma_N} J(F_d)$$

up to isogeny. The desired constructive theorem is an algebraic correspondence

$$\Gamma_N \in \mathrm{CH}^*(F_d \times C_P \times S_T)_{\mathbf{Q}}$$

inducing

$$\boxed{\Gamma_{N*} : e_{\Sigma_N} H^1(F_d) \xrightarrow{\sim} M(C_P) \otimes M(S_T)(1).} \tag{CM4}$$

Once (CM4) is available, its tensor product with the conjugate lower curve packet is a rational $(1,1)$ class on a product of curves or abelian varieties. Lefschetz $(1,1)$ makes it algebraic, and the Shioda–Katsura join maps push the divisor to a codimension-two cycle on X_m^4.

Because the selected character packets are rank one, any nonzero equivariant morphism is an isomorphism. The geometric proof therefore needs only:

1. a Kummer-convolution or half-twist correspondence with the correct character covariance
2. one nonzero period, Jacobi-sum, or intersection matrix coefficient

The remaining issue is constructive, not a missing Hodge-sign classification. Motivic Gauss and Jacobi sum correspondences, including Davenport–Hasse multiplication in Chow-motive form over finite fields, provide a closely related model [11], but the characteristic-zero packet map required here is still to be written.

Explicit CM Packet Correspondence Theorem

For every K23 packet, construct the correspondence (CM4), pair it with the lower curve packet, and push the resulting divisor through the two Fermat join maps. Prove that the final character projection is nonzero.

Chapter 18

Literature interface and constructive novelty

Several established theories meet in this edition.

Shioda–Katsura induction supplies recursive maps among Fermat varieties and explains why curve, surface, and Tate factors are natural constructive atoms [7]. Kang records an all-degree generalized-Hodge existence theorem for Fermat varieties of dimensions three and four [6]. Da Silva studies the reach of explicit Shioda-style packet criteria and isolates degrees where those restricted constructions leave residual packets [10]. Van Geemen and van Geemen–Izadi develop CM half-twists and their realization inside tensor products involving CM abelian varieties and Fermat curves [8, 9]. Otsubo–Yamazaki provide a motivic framework for Gauss and Jacobi sums, including Davenport–Hasse-type identities at the level of Chow motives [11].

The constructive packet program developed here is organized around:

- exact conductor projectors as algebraic graph combinations
- a finite packet certificate grammar
- a worked degree-33 Klein–Delsarte transfer
- infinite old-packet towers and full-conductor promotion
- odd prime-power packet classification
- a weighted dyadic recursion and uniform SAT(3) family
- exact CRT–Kummer reductions for K4 and K23 packets
- a pure-imaginary CM selector and explicit lower Fermat-curve carriers
- an algorithmic packet-resolved Chow correspondence target

Novelty boundary

The all-degree abstract existence statement is background, not the organizing claim of this book. The constructive objective is finer: resolve each rational character orbit, name its local certificate, identify its lower carrier, and write an explicit nonzero correspondence.

Part V. Transfer, Propagation, and Verification Toolkit

Chapter 19

Divisor-lowering transfer and explicit reconstruction

19.1 Degree bookkeeping

Let X have complex dimension n, let $V \subseteq H^{2p}_{\mathrm{prim}}(X, \mathbf{Q})$ be a rational Hodge substructure, and let Y_r be smooth projective. Choose correspondences

$$\Gamma_r \in \mathrm{CH}^{n-p+1}(X \times Y_r)_{\mathbf{Q}}, \qquad \Lambda_r \in \mathrm{CH}^{\dim Y_r + p - 1}(Y_r \times X)_{\mathbf{Q}}.$$

They induce

$$A_r = (\Gamma_r)_* : V \to H^2(Y_r, \mathbf{Q}), \qquad B_r = (\Lambda_r)_* : H^2(Y_r, \mathbf{Q}) \to H^{2p}(X, \mathbf{Q}).$$

Algebraic correspondences preserve rationality and shift Hodge type by the required bidegree.

Theorem 19.1 (Invertible divisor-lowering transfer)**.** *Suppose*

$$T = \sum_{r=1}^{M} B_r A_r$$

preserves V and is invertible on V. Then every rational Hodge class in V is algebraic.

Proof. Let $\alpha \in V$ be of type (p, p). Each $A_r\alpha$ is rational of type $(1, 1)$ on Y_r, hence is the class of a rational divisor D_r by the Lefschetz $(1, 1)$ theorem. Therefore

$$T\alpha = \sum_r (\Lambda_r)_*[D_r]$$

is algebraic on X.

Let $\chi_T(t) = t^d + c_{d-1}t^{d-1} + \cdots + c_0$ be the characteristic polynomial on V. Invertibility means $c_0 \neq 0$. Cayley-Hamilton gives

$$T^{-1} = -\frac{1}{c_0}(T^{d-1} + c_{d-1}T^{d-2} + \cdots + c_1 I).$$

Each power of T is induced by compositions and rational sums of algebraic correspondences. Thus T^{-1} preserves algebraic cycle classes. Hence $\alpha = T^{-1}(T\alpha)$ is algebraic. □

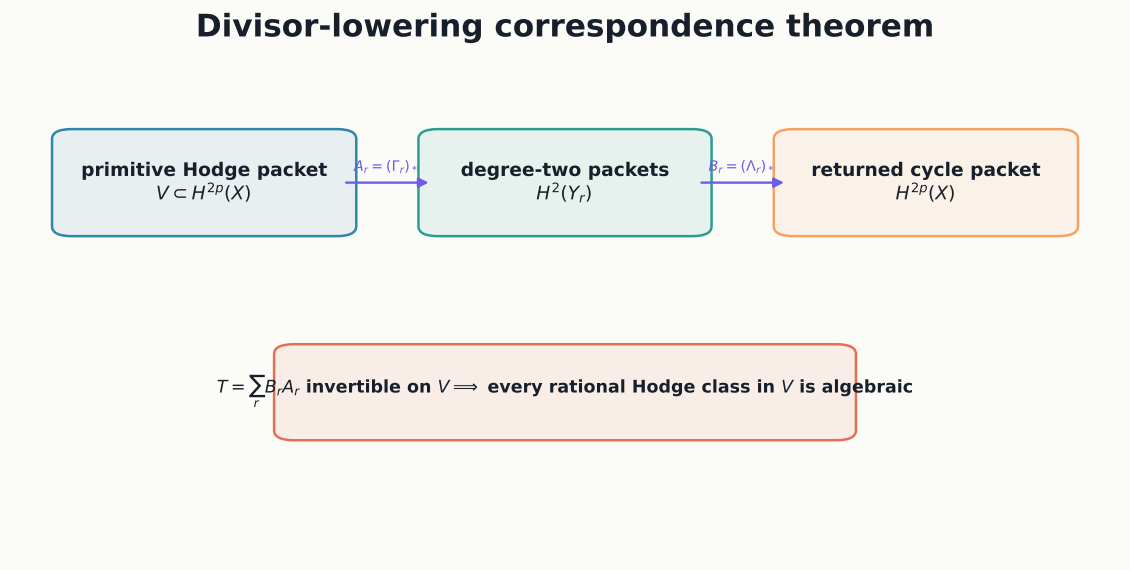


Figure 19.1: A primitive Hodge class is lowered to divisor degree, algebraized there, and returned through an invertible transfer.

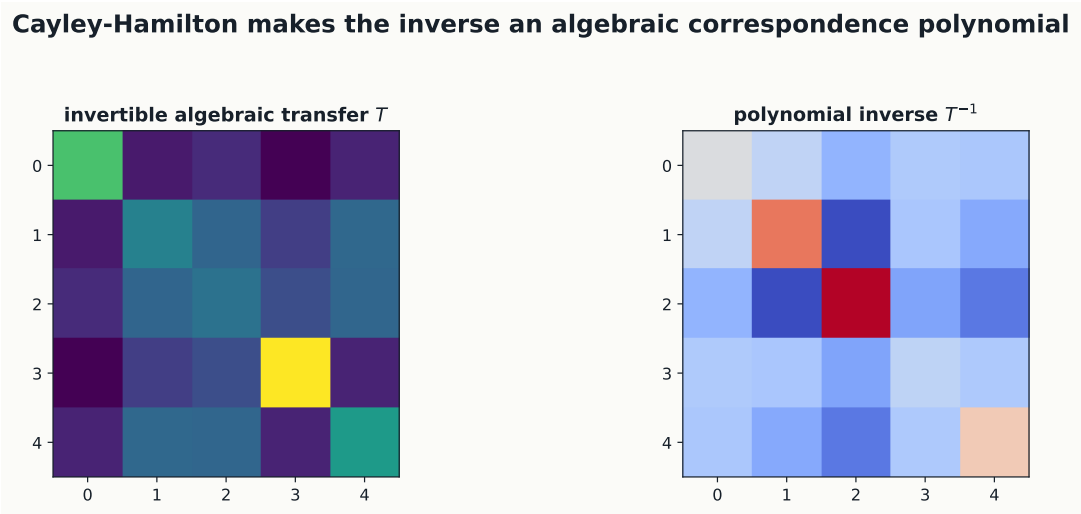


Figure 19.2: The inverse transfer is a rational polynomial in the original algebraic correspondence operator.

19.2 Positive multi-channel form

Assume the polarized adjoint $A_r^\dagger$ is realized by an algebraic return correspondence B_r. This is an additional geometric hypothesis, not an automatic consequence in arbitrary settings. Then

$$T = \sum_r A_r^\dagger A_r$$

is positive and

$$\langle v, Tv \rangle = \sum_r \|A_r v\|^2.$$

Therefore

$$\ker T = \bigcap_r \ker A_r.$$

Joint injectivity of the lowering channels is enough.

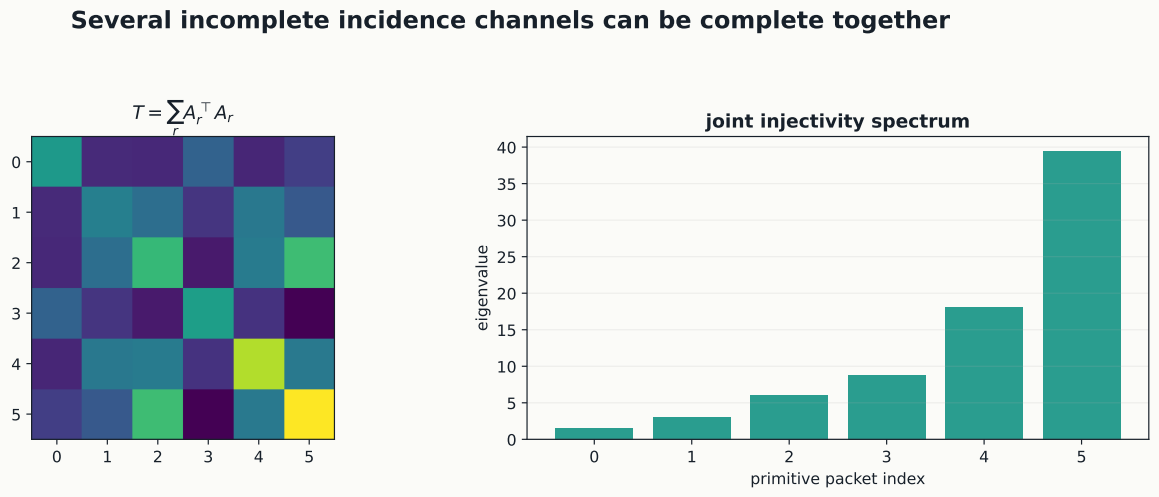


Figure 19.3: Several incidence channels may be individually incomplete while their positive sum is invertible.

Corollary 19.2 (Finite Gram certificate)**.** *Choose a rational basis $e_1, \dots, e_d$ of V. If*

$$G_{ij} = \sum_r \langle A_r e_i, A_r e_j \rangle_{Y_r}$$

has nonzero determinant, every rational Hodge class in V is algebraic.

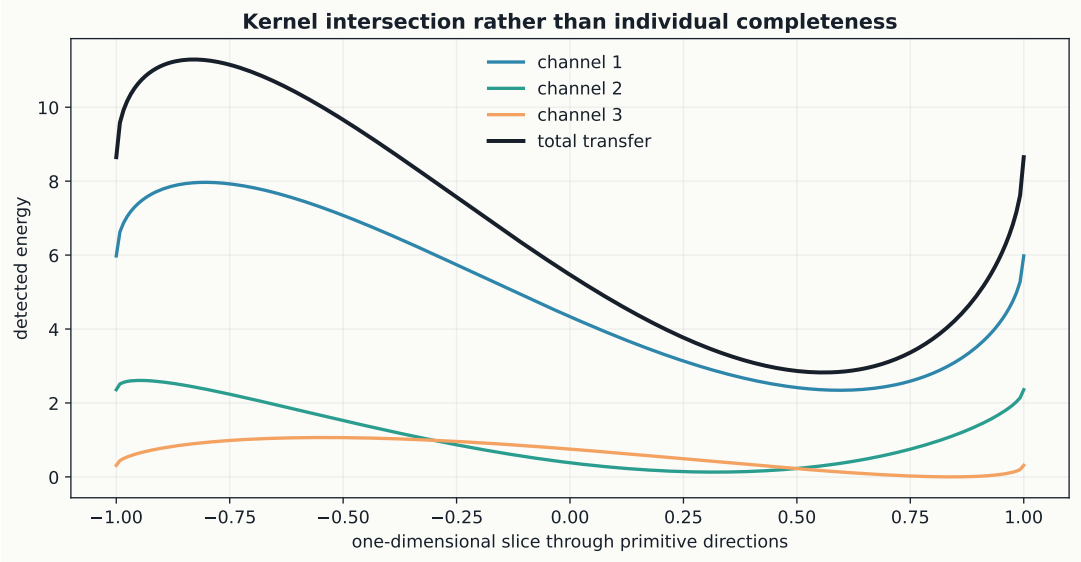


Figure 19.4: The correct criterion is the vanishing of the joint kernel, not completeness of one channel.

Chapter 20

Propagation through connected families

Let $\pi : \mathcal{X} \to S$ and $\rho_r : \mathcal{Y}_r \to S$ be smooth projective families over a connected base. Relative algebraic correspondences induce morphisms of Gauss-Manin local systems. In a flat rational trivialization, their transfer matrix is constant.

Theorem 20.1 (Propagation of invertibility)**.** *If a relative transfer T_s is invertible on one fiber s_0, it is invertible on every fiber in the connected base on which the local systems and correspondences are defined.*

Proof. The determinant of the constant rational matrix representing the local-system morphism is locally constant. It is nonzero at s_0, hence everywhere on the connected base. □

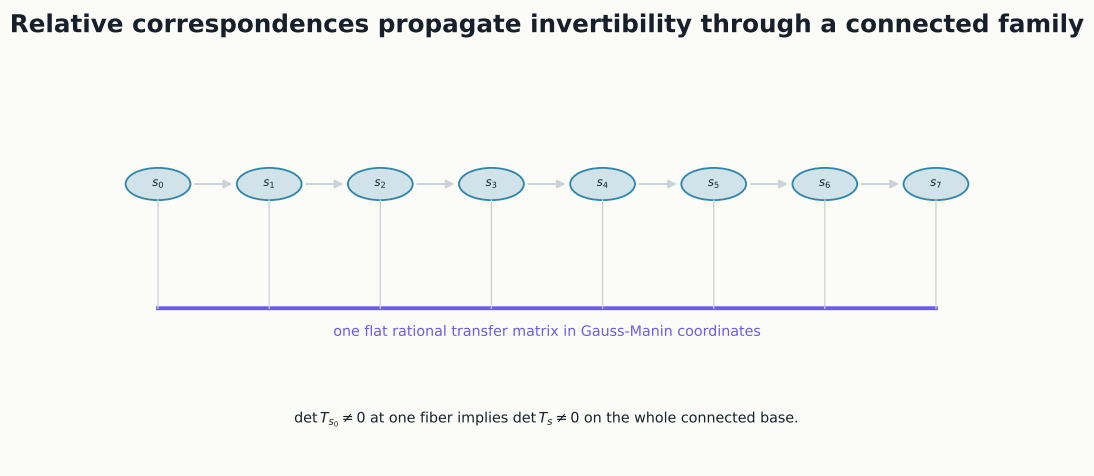


Figure 20.1: One verified fiber can control a connected smooth family when the correspondence is relative and flat.

Chapter 21

Verification and formalization workflow

Numerical experiments can scout candidate families but cannot replace an exact rank certificate. The intended pipeline is:

$$\text{exact incidence data} \longrightarrow \text{rational cohomology basis} \longrightarrow \text{packet transfer matrices} \longrightarrow \text{rank or positivity certificate} \longrightarrow \text{symbolic theorem proof.}$$

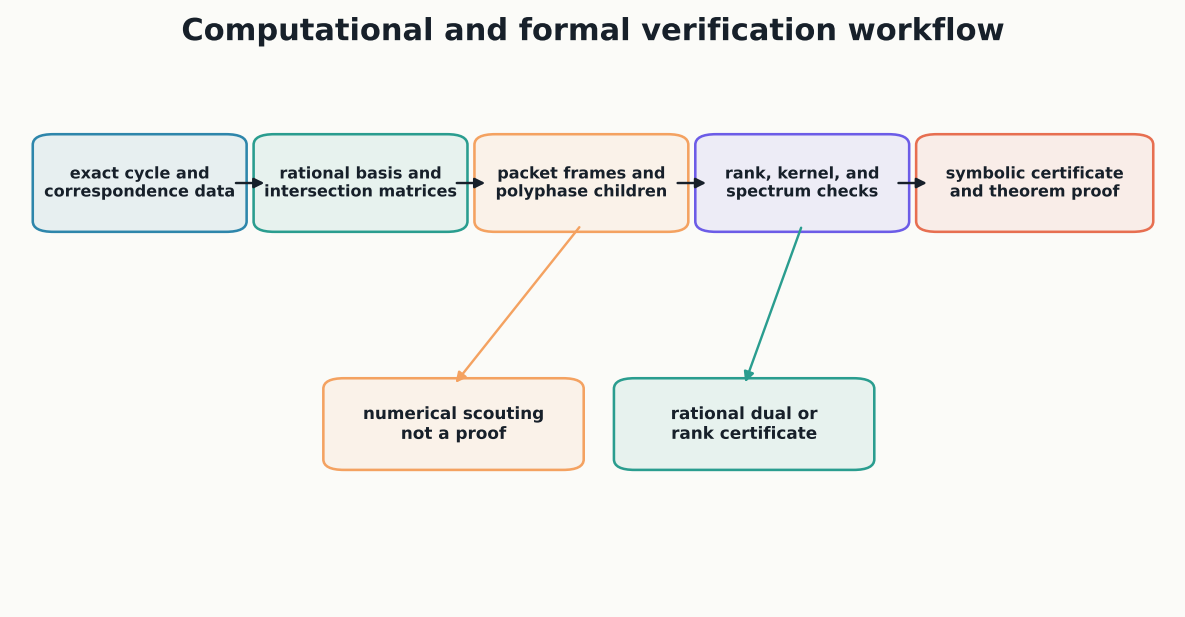


Figure 21.1: Numerical scouting is separated from rational certification and formal theorem verification.

21.1 Reproducible finite checks

The accompanying Python generator produces all figures and runs eighteen independent finite checks covering:

- frame idempotency and range rank
- positive Hodge Gram spectrum

- exact modulo-2 and modulo-4 polyphase assembly
- positive transfer rank and energy identities
- Cayley-Hamilton inversion
- contraction of the removable nonlinear packet
- Adams projector idempotency and degree isolation
- nilpotent trace blindness
- support multiplicity recovery
- finite rank stabilization

These tests verify the finite examples and implementation. They do not establish the unwritten universal correspondence theorem.

Part VI. Verification and Constructive Completion

Chapter 22

Verification ledger

The complete edition separates symbolic theorems, finite exact checks, geometric correspondences, and literature claims.

22.1 Finite checks

The base projector-calculus release passed 18 finite checks. The Fermat extension adds:

- 14 exact degree-33 Klein checks
- 4 infinite tower checks
- 4 exact conductor-polyphase checks
- 13 odd prime-power checks
- 89,478,480 dyadic weighted coefficient identities
- exact packet classifications at dyadic degrees through 128
- exact balanced-packet audit through degree 100
- 364 K23 CM-type certificates

22.2 What finite checks establish

They establish exact arithmetic identities, matrix products, orbit counts, conductor decompositions, Hodge balance, and nonnegative multiset equalities for every tested instance. They also detect false universal pattern claims quickly.

22.3 What finite checks do not establish

They do not replace:

- a universal dyadic uniqueness proof
- a universal mixed-prime phase-classification proof beyond the audited range

- a Chow-level Kummer–Schur or half-twist correspondence
- a historical priority audit

Chapter 23

Constructive completion theorem stack

The constructive Fermat program can be stated as a dependency stack.

Theorem 23.1 (Conditional packet-construction theorem)**.** *Assume the following correspondence statements:*

1. *saturated Fermat joins are associative on Chow motives and complementary Tate factors cancel*
2. *every Delsarte torus map used by the certificate grammar extends through a resolved graph with full-support boundary annihilation*
3. *the weighted dyadic classification has no residual beyond the uniform SAT(3) family*
4. *every K5, K4, and K23 local phase packet has its prescribed Kummer–Schur or CM half-twist Chow correspondence*

Then every rational middle character packet on every Fermat fourfold has an explicit packet-resolved algebraic cycle representative.

Proof. Apply the exact conductor projectors. Descend each nonzero component to its full-conductor seed. Classify the seed by the pair, saturation, curve, Delsarte, dyadic, or Kummer phase grammar. Use the corresponding Chow map to produce a nonzero cycle projection. Pull old packets back to the original degree and sum the orthogonal conductor components. Finally sum the Galois orbit to obtain a rational cycle. □

Unconditional constructive completion

Write the missing correspondence maps explicitly and verify their nonzero packet matrix coefficients. The most concentrated target is the rank-one CM map (CM4).

Chapter 24

Research checklist

Task	Status	Required proof object
Universal balanced packet equation	Complete	Finite residue matrix
Exact conductor descent	Complete	Coordinate-power graph
Möbius conductor projectors	Complete	Rational graph combination
Degree-33 Klein packet	Manuscript-level complete	Resolved graph and residue pullback
Infinite old-packet towers	Complete	Pullback/pushforward identity
Odd prime-power classification	Symbolically complete	Primitive-character filtration
Dyadic weighted recursion	Identity complete	Uniform uniqueness induction
K4 CRT reduction	Complete	Lower balanced four-packet
K23 local phase classification	Complete	Pair–triple lemma
K23 CM sign descent	Complete	Imaginary selector Δ_ℓ
Explicit lower CM carrier	Identified	Fermat-curve character idempotent
CM carrier Chow map	Open	Nonzero algebraic correspondence
All-degree constructive algorithm	Conditional	Assemble all correspondence channels
Constructive literature map	Open audit	Separate published existence results from explicit correspondence priority

Chapter 25

Constructive conclusion

The Fermat-fourfold program is now organized as a finite hierarchy of character addresses and explicit transfers. A packet begins as a balanced six-tuple. Exact conductor projectors descend it to its minimal modulus. Prime-local analysis then selects the relevant grammar: complementary pairs, saturated surface joins, curve factors, Delsarte descent, weighted dyadic refinement, or Kummer–Schur phase decomposition.

The degree-33 packet demonstrates why a constructive theory must allow cross-family transfers. It escapes the obvious lower-packet grammar but is reached by a monomial map to a smooth cubic fourfold. The prime-power classification shows that odd primes and the dyadic prime have genuinely different packet geometries. The mod-5 audit compresses mixed-prime residuals into K5, K4, and K23 local types. The K23 identity isolates a weight-one factor, and the pure-imaginary cyclotomic element

$$\Delta_\ell = (1-\zeta_q^c)(1-\zeta_q^d)(1-\zeta_q^e)$$

selects the upper/lower CM embedding sectors exactly.

The carrier-selection problem is therefore explicit: the half-twist lies in a character factor of the Jacobian of a lower Fermat curve. The remaining constructive task is rank one.

Final constructive endpoint

Construct one nonzero equivariant Chow correspondence

$$e_\Sigma H^1(F_d) \longrightarrow M(C_P) \otimes M(S_T)(1)$$

for each K23 carrier, then pair it with the conjugate lower curve packet and push the resulting divisor through the Fermat join maps.

This endpoint is stronger than an abstract existence statement and more informative than a finite classification. It asks for an algorithmic cycle formula that reveals how conductor, local phase, imaginary CM type, and algebraic transfer fit together.

Appendix A

Formula registry

Object	Formula
Fermat fourfold	$X_m^4 = \{x_0^m + \cdots + x_5^m = 0\} \subset \mathbf{P}^5$
Hodge level address	$\ell_a(u) = m^{-1} \sum_i \langle u a_i \rangle_m$
Balanced packet space	$\mathcal{B}_m = \{n \in \mathbf{N}^{m-1} : \mathbf{1}^T n = 6,\ R_m n = 3m\mathbf{1}\}$
Exact conductor	$c(a) = m/\gcd(m, a_0, \ldots, a_5)$
Old-space projector	$E_{m,d} = (m/d)^{-5} \pi_{m/d}^* \pi_{m/d*}$
Conductor projector	$Q_{m,c} = \sum_{d\mid c} \mu(c/d) E_{m,d}$
Complementary pair	$p_r = e_r + e_{m-r}$
Balanced four-packets	$\mathcal{F}_m = \{\gamma : \mathbf{1}^T \gamma = 4,\ R_m \gamma = 2m\mathbf{1}\}$
Saturated certificate	$n + \sum_{j=1}^{2N-3} p_{r_j} = \sum_{i=1}^{N} \gamma_i$
Positive cycle frame	$S = CC^* = \sum_j \lvert c_j\rangle\langle c_j\rvert$
Packet coverage	$\lambda_{\min}(H^* S H) > 0$
Positive transfer	$T = \sum_r A_r^\dagger A_r$
Degree-33 matrix identity	$AC = 33 I_6$
Five-primary family	$\alpha_{5q} = (1, 1+q, 1+2q, 1+3q, 1+4q, 5q-5)$
Dyadic low/high packets	$L_s n(b) = n(b) + n(b+2^{s-1})$, $D_s n(b) = n(b) - n(b+2^{s-1})$
K4 lower packet	$\Gamma_{b;c,d} = (b, c, d, -5^{-1}b)$ modulo q
K23 identity	$A + (5c, m-5c) + (5d, m-5d) = C_P + S_T + L$
Imaginary selector	$\Delta_\ell = (1-\zeta_q^c)(1-\zeta_q^d)(1-\zeta_q^e)$
CM type	$\Sigma_N = \{u : \operatorname{Im} \sigma_u(\Delta_\ell) > 0\}$
Complex structure	$J = iP_{\Sigma_N} - iP_{\overline{\Sigma_N}}$
Explicit carrier	$A_{\Sigma_N} = e_{\Sigma_N} J(F_d)$ up to isogeny
Rank-one Chow target	$e_{\Sigma_N} H^1(F_d) \to M(C_P) \otimes M(S_T)(1)$

Appendix B

Correspondence degree and twist bookkeeping

Let X have dimension n and Y dimension m. A correspondence $\Gamma \in \mathrm{CH}^r(X \times Y)$ acts by

$$\Gamma_* : H^i(X) \to H^{i+2(r-n)}(Y).$$

To lower $H^{2p}(X)$ to $H^2(Y)$, choose $r = n - p + 1$. To return $H^2(Y)$ to $H^{2p}(X)$, choose a correspondence of codimension $m + p - 1$ on $Y \times X$. This bookkeeping is essential when inserting polarizations into transpose incidence correspondences.

Appendix C

Verification tables

The finite tables below summarize named test suites, dyadic packet classification, odd prime-power spot checks, residual degrees through 100, and the K23 CM audit. They are exact certificates for the displayed instances and reproducible code paths.

C.1 Named suites

Suite	Passed	Total
projector resolved hodge calculus	18	18
fermat33 klein closure	14	14
infinite klein tower	4	4
conductor polyphase	4	4
odd prime power audit	13	13
Total	53	53

C.2 Dyadic classification

m	Full orbits	Pair	SAT(2)	SAT(3)	Unresolved
8	13	9	4	0	0
16	45	27	18	0	0
32	134	91	42	1	0
64	419	331	87	1	0
128	1435	1259	175	1	0

C.3 Odd prime-power spot checks

m	Full orbits	Pair/SAT(2)	Residual
49	124	124	0
81	436	436	0
125	827	826 pair	1

C.4 Residual degrees through 100

Degree	Post-saturation representatives
25	`(1,6,11,16,20,21)`
33	`(1,4,16,22,25,31)`
35	`(1,8,15,22,29,30)`
39	`(1,7,16,22,34,37)`
40	`(1,9,17,25,33,35)`
45	`(1,10,19,28,37,40); (1,19,20,28,30,37)`
50	`(1,7,27,30,41,44); (1,11,21,31,41,45); (1,13,22,32,40,42)`
55	`(1,12,23,34,45,50)`
65	`(1,14,27,40,53,60)`
70	`(1,6,43,45,57,58); (1,15,29,43,57,65); (1,20,24,42,61,62); (1,20,29,43,57,60); (1,29,35,43,45,57)`
75	`(1,16,31,46,61,70)`
80	`(1,17,33,49,65,75)`
85	`(1,18,35,52,69,80)`
90	`(1,8,26,62,85,88); (1,19,37,55,73,85); (1,19,37,60,73,80); (1,19,37,65,73,75); (1,23,38,55,74,79); (1,38,46,55,56,74); (2,38,45,55,56,74)`
95	`(1,20,39,58,77,90)`
100	`(1,21,41,61,81,95)`

C.5 CM half-twist audit

Item	Value
K23 rational orbits	7
Exact K23 certificates	364
Opposite curve CM sign	True
Lower-conductor factorization	True
Upper-half-plane CM translate	True

Appendix D

Source and claim map

The edition uses four classes of input.

1. The internal projector-resolved seed calculus supplies positive packet assembly, exact parent-child refinement, weighted dyadic transfer, and obstruction bookkeeping [3, 4, 5].
2. Classical Fermat geometry supplies diagonal character decompositions, Shioda–Katsura joins, Fermat-curve factors, and coordinate-power maps [7].
3. Established correspondence theory supplies the cubic-fourfold incidence engine and CM half-twist background [2, 8, 9].
4. Motivic Gauss and Jacobi sum identities supply the natural algebraic model for Kummer and Davenport–Hasse packet relations [11].

The manuscript's exact internal statements are the finite packet equations, conductor projectors, packet identities, prime-power classifications, weighted dyadic identities, CRT–Kummer reductions, and imaginary CM selector. Geometric statements that require a resolved graph, an associative join in Chow motives, or a nonzero equivariant correspondence are marked accordingly. Historical priority for each explicit correspondence should be audited separately from the mathematical proof.

Afterword on constructive status

The edition contains three kinds of statement. First are exact symbolic theorems and finite certificates: residue equations, conductor decompositions, packet identities, matrix products, and CM sign laws. Second are correspondence schemas whose hypotheses explicitly require algebraic maps, resolved graphs, or Chow-motive joins. Third is the remaining constructive target: a nonzero rank-one CM packet correspondence. These layers should remain distinct when the work is cited, formalized, or prepared for publication.

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